TPP-DLW Toolpath Generator

New User Guide for STL-to-TXT Conversion

This guide explains how to convert an STL file into the tab-separated segment text format used by the LabVIEW writing interface. It focuses on the fastest current workflow and the checks that prevent false toolpath lines.

Recommended path: use `heightmap_raster_export.m` for pixel-exact height-map STL/CSV files. Use the modular `tppdlw_process` pipeline for general STL/STEP workflows or the GUI.

1. Project Layout

Height-map raster	`heightmap_raster_export.m` - direct height-map STL/CSV to TXT exporter.
Contour STL exporter	`stl_slice_export_mm_woodpile_true_resample.m` - contour-slicing STL exporter for non-height-map cases.
General pipeline	`tppdlw_process.m` plus `core/` helpers for STL/STEP imports and arbitrary scan angles.
Preview tools	`viz/preview_toolpath.m`, `viz/preview_3d.m`, and layer comparison tools.
Examples/tests	`examples/` and `tests/` show basic usage and expected output format.
Generated data	`output/`, STL files, and large TXT exports are ignored by Git by default.

2. Quick Start

From MATLAB, open the project folder and run:

```
cd('/path/to/3D_Printer_construct')
run('heightmap_raster_export.m')
```

- Set `HeightMapPath` to your input height-map STL or CSV file.
- Set `OutTxt` to the TXT file you want to create.
- Set `BaseHeight` if the source CSV needs a support base; leave it at 0 if the STL already includes one.
- Start with coarse `XYPitch` and `DZ` for preview runs, then restore final values for production.
- Read the console summary. Height-map raster export should not report odd contour-crossing warnings.

3. Output Format

The output TXT is tab-separated, with one row per laser-on scan segment or Z transition:

```
X1	Y1	Z1	X2	Y2	Z2
```

- Rows where `Z1 == Z2` are scan/write segments.
- Rows where `Z1 != Z2` are layer-to-layer Z moves.
- Coordinates are in millimeters.
- The height-map raster exporter writes positive build height as negative stage Z when `StageZConvention` is true.

4. Main Parameters

Parameter	Purpose	Typical use
`HeightMapPath`	Input height-map STL or CSV file.	Use this path for pixel-exact height maps.
`OutTxt`	Destination tab-separated TXT file.	Use `output/name.txt` for generated files.
`TargetMaxXY`	Scales the model to a target XY footprint in mm.	Scalar for square fit, or [maxX maxY].
`XYPitch`	Spacing between adjacent raster scanlines.	Larger for preview, final value for production.
`DZ`	Layer thickness / vertical slicing step.	Larger for preview, final value for production.
`PixelPitch`	Source pixel pitch for CSV input.	STL can read this from the generated header.
`BaseHeight`	Support base height in source units.	Use 0 for STL that already includes a base; use 0.5 for a micron-valued CSV needing a 0.5 um base.
`WoodpileMode`	Alternates horizontal and vertical writing by layer.	Keep true for orthogonal resampling.
`Serpentine`	Alternates scan direction to reduce travel.	Usually true.
`CoordMode`	Uses voxel edges or grid centers for endpoints.	Use edges for full-width base rows.
`StageZConvention`	Writes positive build height as negative stage Z.	Usually true for the printer stage convention.
`OutputSignificantDigits`	Controls compact TXT numeric precision.	Default 6 keeps values like 0.0002 readable.
`Tolerance_mm`	Height comparison tolerance in mm.	Default is usually fine.

5. Size and Pixel Pitch

- `PixelPitch` describes the original height-map pixel spacing in source units, such as 6 for a 6 um CSV pixel.
- `TargetMaxXY` controls the final maximum XY footprint in millimeters.
- `XYPitch` is the printer raster scanline spacing in millimeters. It controls write density and TXT size, not the source model footprint.
- If `TargetMaxXY` is set, the final source-pixel pitch is derived from target size and grid count. For a 335 x 335 map at 1.005 mm, one source pixel becomes 1.005 / 335 = 0.003 mm.
- `BaseHeight` is added in `heightmap_to_segments` before raster scanlines are generated, so it creates full support-base scanlines across the height-map footprint.

6. Preview and Quality Checks

After generating a TXT file, preview a layer before printing:

```
addpath(genpath(pwd))
segments = load('output/Final.txt');
preview_toolpath(segments, 'Layer', 1, 'Mode', '2d')
preview_3d(segments, 'EveryN', 2)
```

Before sending a file to the printer, check for long strokes that cross empty space, out-of-bounds coordinates, unexpected zero-length rows, and unusually high segment counts on one layer.

If the first rows are short fragments along a boundary such as X = 0, a contour-slicing workflow is being used. Use heightmap_raster_export.m for pixel-exact height-map data so the file starts with hatch-fill rows.

Important: the contour-slicing fallback skips unresolved odd scanlines instead of force-pairing them. For pixel-exact height maps, prefer the raster workflow so contour crossings are not used at all.

7. Troubleshooting

Long lines across empty regions	Usually caused by using contour slicing on height-map data. Use height-map raster export first.
Very slow conversion	Increase `XYPitch` and `DZ` for preview.
Missing small features	Check whether `XYPitch` or `DZ` is too coarse.
TXT is huge	This is expected for fine pitch and large footprints. Keep generated TXT files under `output/` so Git ignores them.
STL import fails	Confirm the STL is binary or valid ASCII, not truncated, and that the file path is correct.

8. Recommended Workflow

- **Preview pass:** use coarse pitch and layer height; generate quickly and inspect several layers.
- **Geometry check:** confirm the recovered height-map pitch, base height, and output span match the intended print.
- **Final pass:** restore production `XYPitch` and `DZ`; regenerate TXT.
- **Visual QA:** preview first, middle, and last layers; check that no scanline crosses empty space.
- **Archive:** commit code/config changes, not large generated files. Keep STL/TXT outputs ignored unless you intentionally need to version them.

9. Quick Checklist Before Printing

[ ] Height-map source is correct	Right file, right units, expected base height and bounding box.
[ ] Parameters reviewed	Target size, XY pitch, DZ, coordinate mode, and output path are intentional.
[ ] Console warnings reviewed	No unexpected truncated STL, missing height-map cells, or out-of-bounds warnings.
[ ] Layers previewed	First, middle, and final layers look physically plausible.
[ ] Output format checked	TXT has six numeric columns and the expected number of layers.

[ ] Generated files managed	Large STL/TXT files are kept out of Git unless deliberately tracked.

For maintainers: keep the false-line prevention behavior conservative. Skipping an unresolved scanline is preferable to writing a long segment across empty space.